

Sri Lanka Institute of Information Technology
Faculty of Computing | Department of Computer Science

IT3021: Data Warehousing and Business Intelligence

Assignment 2

SSAS Cube, OLAP Operations & Power BI Reports

Field	Details
Course Code	IT3021 - Data Warehousing and Business Intelligence
Year / Semester	Year 3 / Semester 1
Assignment	Assignment 2
Name	Hewage Y.U
ID	IT23744820
Submission Date	April 2026
Mode	Take-home
Max Marks	100 marks (20% of Final Grade)

Chapter 1: Data Source for Assignment 2

1.1 Overview

The data source for Assignment 2 is the Wish_Ecommerce_DW data warehouse implemented and populated in Assignment 1. This data warehouse is hosted on SQL Server 2019 Express Edition on the local instance HEWAGE2003\SQLEXPRESS and follows a Star Schema dimensional model built from the Wish Summer Products and Sales dataset sourced from Kaggle.

The data warehouse contains transactional product and sales data from the Wish e-commerce marketplace covering product listings, merchant information, pricing, ratings, and sales volumes.

1.2 Data Warehouse Schema

The data warehouse implements a Star Schema design consisting of one central fact table and three dimension tables:

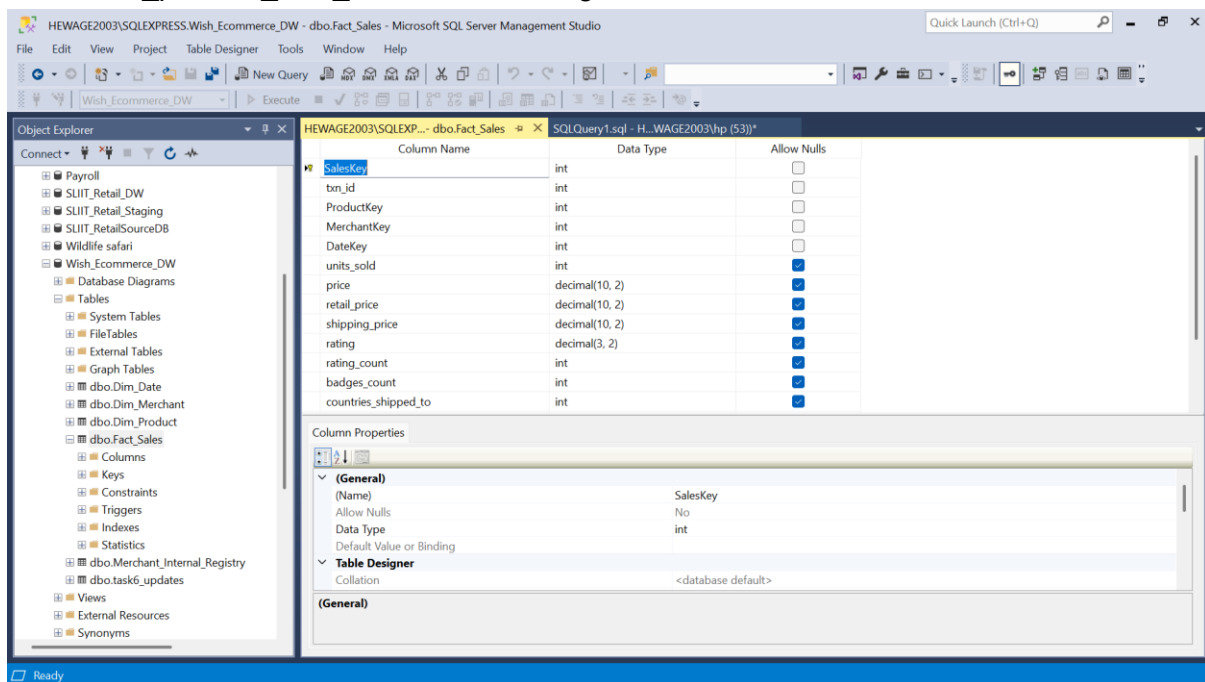
Table	Type	Rows	Description
Fact_Sales	Fact Table	4,719	Central fact table containing sales transactions with measures
Dim_Product	Dimension (SCD Type 1)	~3,122	Product master data including title, price, color, origin
Dim_Merchant	Dimension (SCD Type 2)	~5,748	Merchant master data with historical tracking
Dim_Date	Date Dimension	2,922	Calendar dimension from 2019-01-01 to 2026-12-31



1.3 Fact Table Measures

The Fact_Sales table contains the following measurable facts used in the SSAS cube and Power BI reports:

- units_sold - Number of units sold per product
- price - Current selling price (EUR)
- retail_price - Original retail price before discount
- shipping_price - Cost of shipping to buyer
- rating - Average customer rating (1-5)
- rating_count - Total number of customer ratings
- badges_count - Number of quality badges earned
- accm_txn_create_time - Transaction creation timestamp
- accm_txn_complete_time - Transaction completion timestamp
- txn_process_time_hours - Processing time in hours



Chapter 2: SSAS Cube Implementation

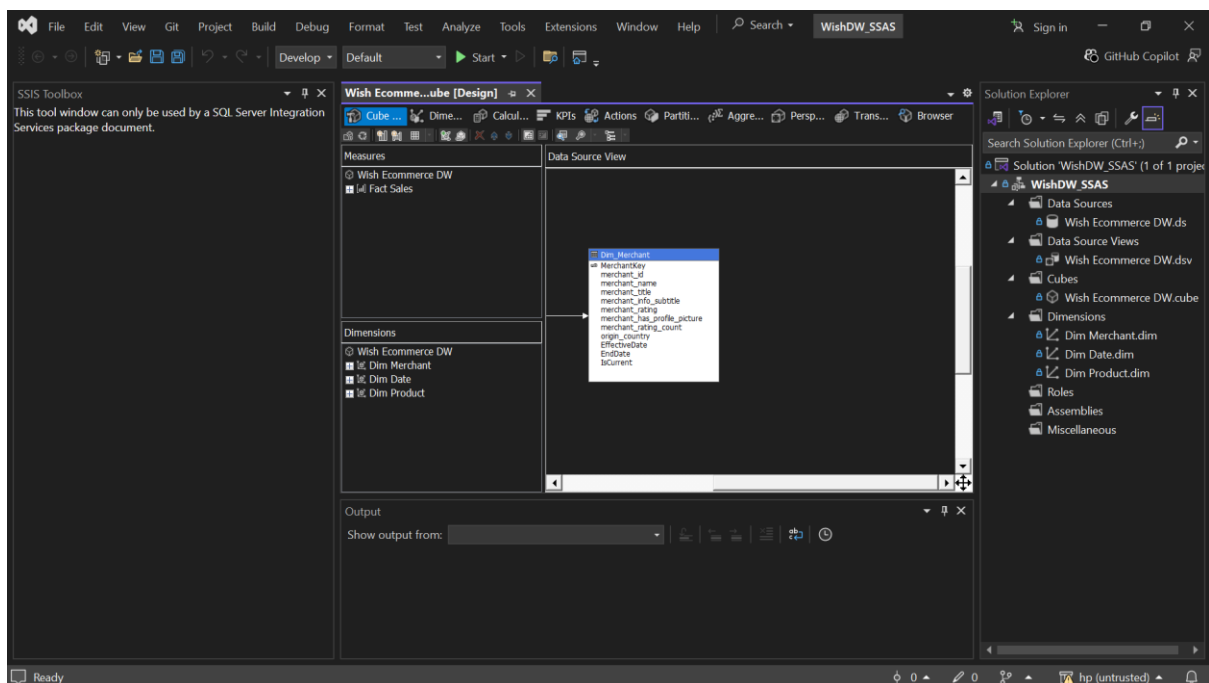
2.1 Overview

An SQL Server Analysis Services (SSAS) Multidimensional cube was created using Visual Studio 2022 with the Microsoft Analysis Services Projects 2022+ extension. The cube is deployed to the SSAS instance HEWAGE2003\SSAD and uses Wish_Ecommerce_DW as its data source.

2.2 SSAS Project Setup

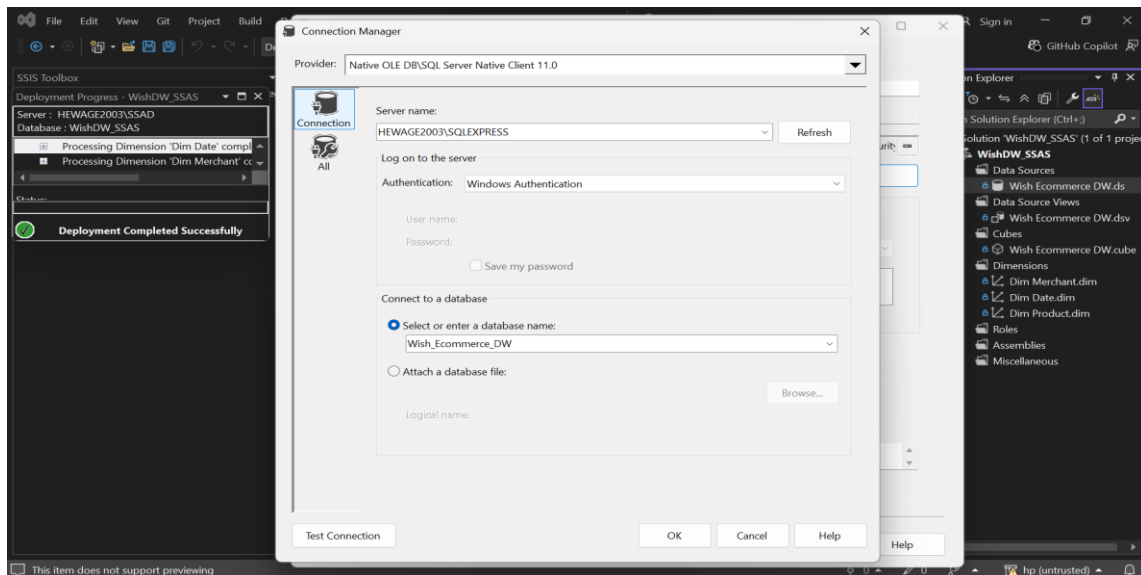
Step 1: Create New SSAS Project

A new Analysis Services Multidimensional Project was created in Visual Studio 2022 named WishDW_SSAS. The project template was installed via the Microsoft Analysis Services Projects 2022+ extension from the Visual Studio Marketplace.

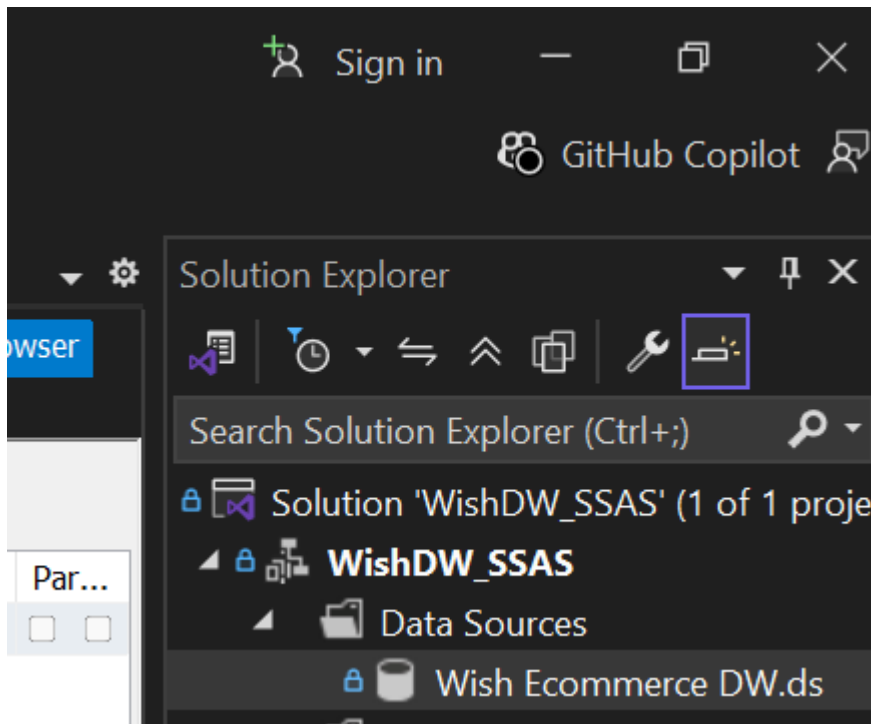


Step 2: Create Data Source

A new Data Source was created by right-clicking the Data Sources folder in Solution Explorer and selecting New Data Source. The connection was configured to connect to HEWAGE2003\SQLEXPRESS and the Wish_Ecommerce_DW database using Windows Authentication.



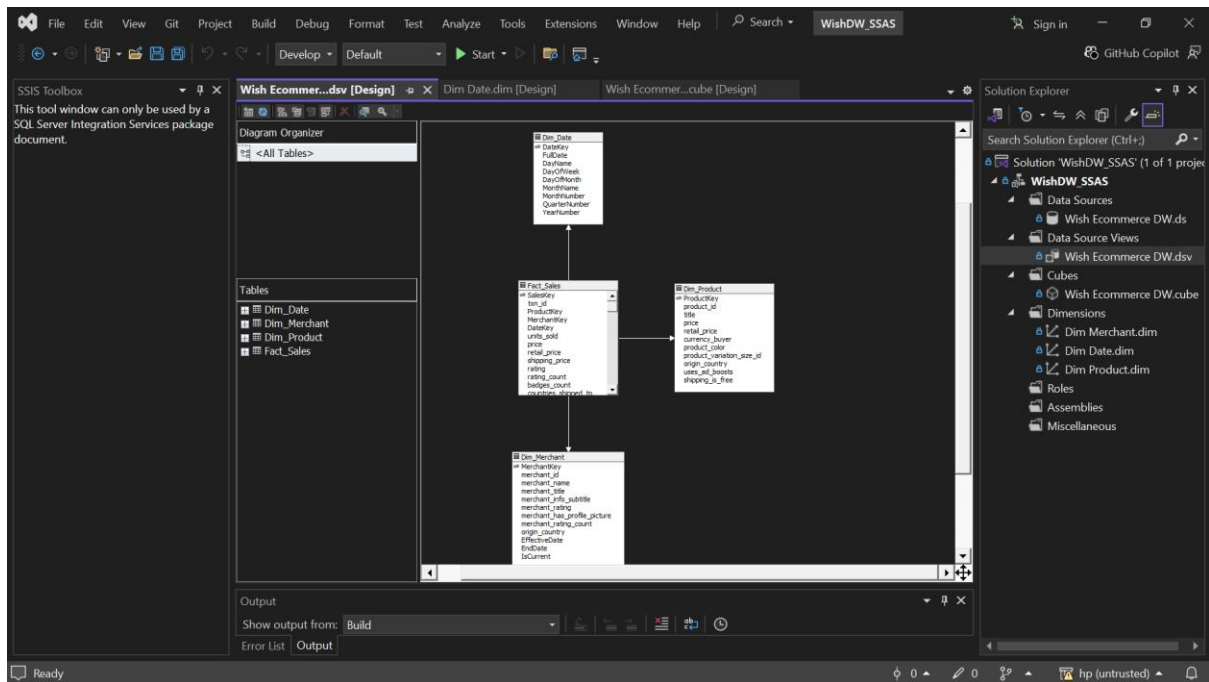
The NT SERVICE\MSOLAP\$SSAD service account was granted db_datareader access to Wish_Ecommerce_DW in SQL Server to allow the SSAS service to process the cube data.



Step 3: Create Data

Source View

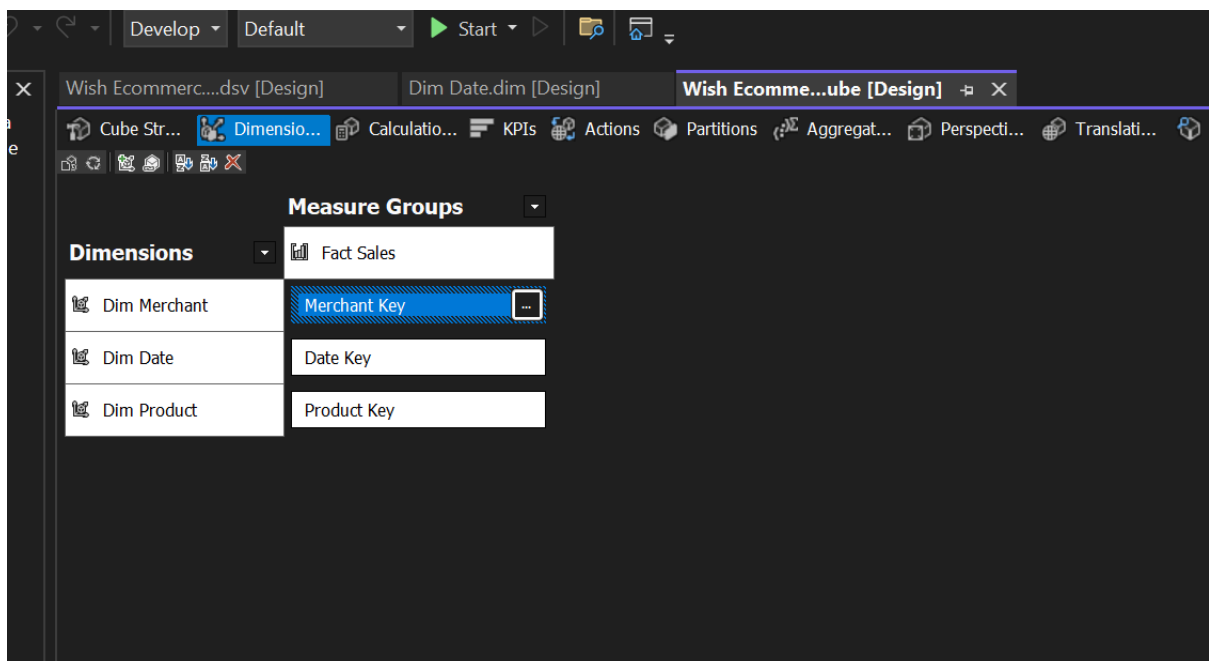
A Data Source View was created including the four tables: Fact_Sales, Dim_Product, Dim_Merchant, and Dim_Date. The view displays the relationships between the fact table and dimension tables.



2.3 Cube Design

Step 4: Create the Cube

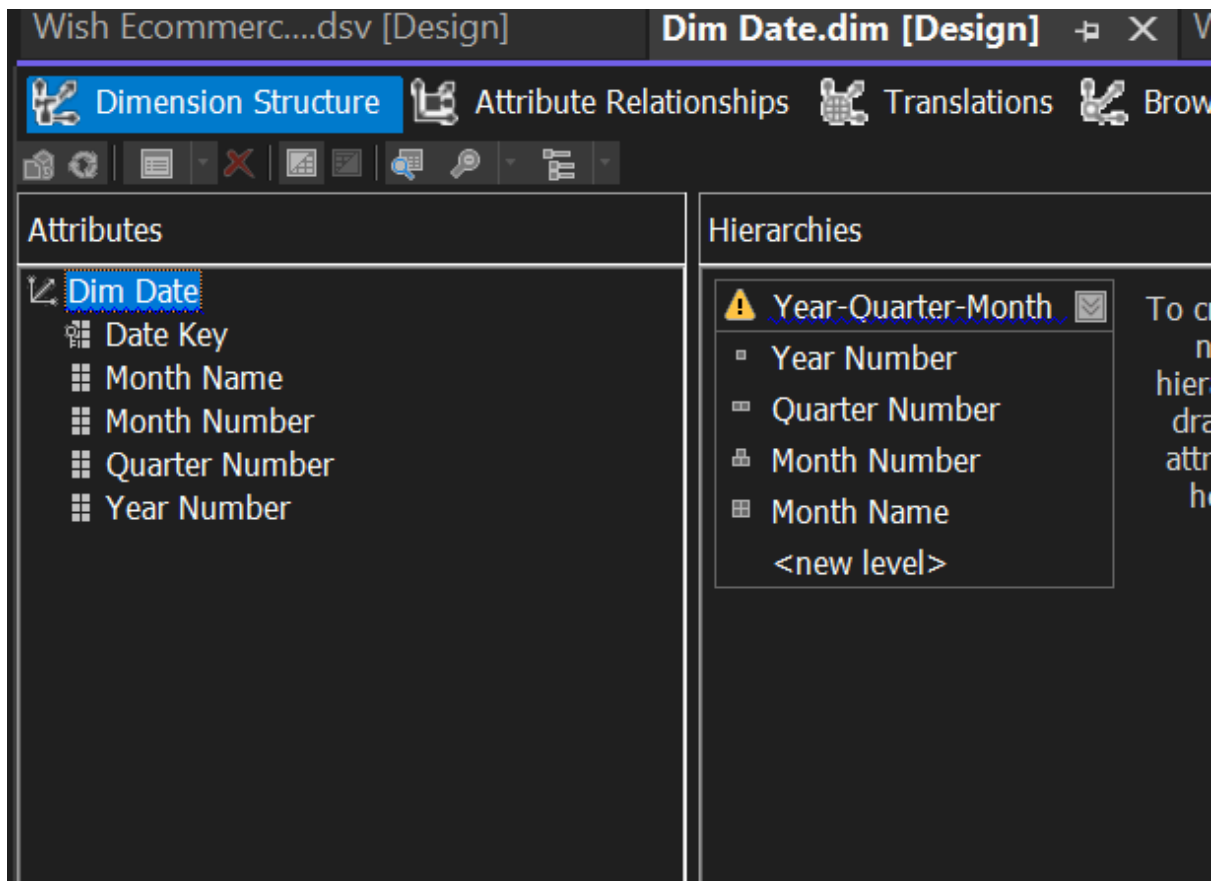
The cube was created using the Cube Wizard selecting Fact_Sales as the fact table, with all key measures included and all three dimensions connected.



2.4 Hierarchy Implementation

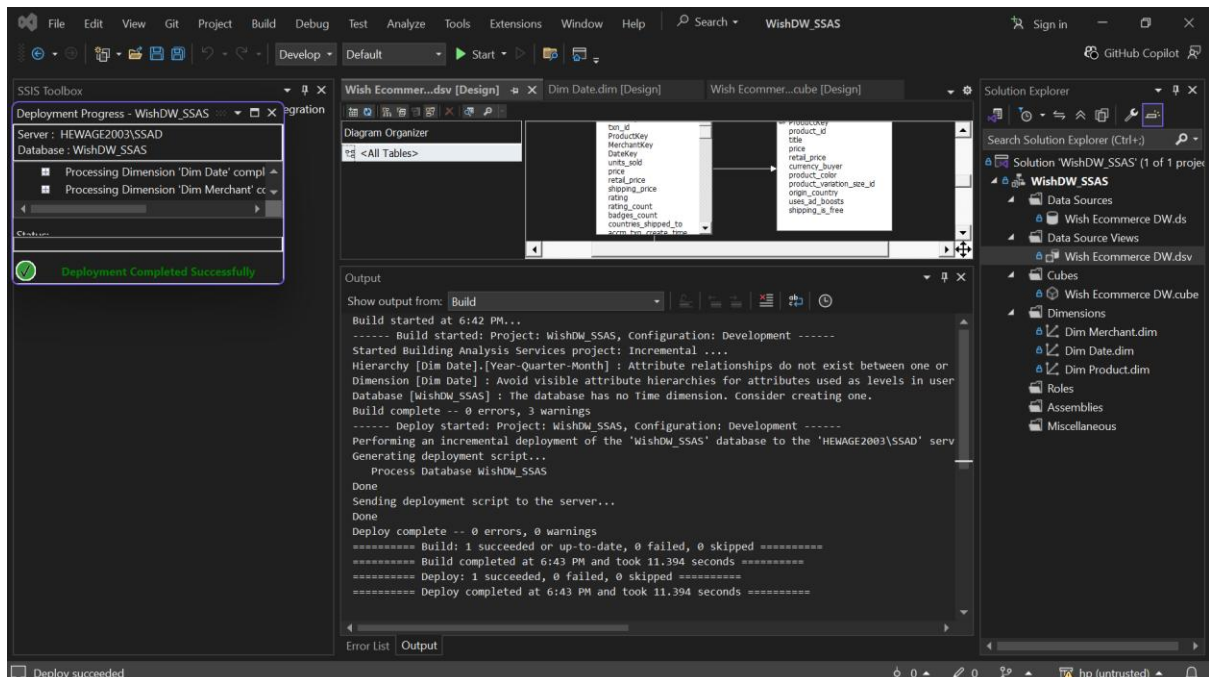
Step 5: Date Hierarchy - Year-Quarter-Month

A date hierarchy was implemented in the Dim_Date dimension. The attributes YearNumber, QuarterNumber, MonthNumber, and MonthName were first added to the Attributes panel by dragging them from the Data Source View. They were then combined into a single user-defined hierarchy named Year-Quarter-Month.

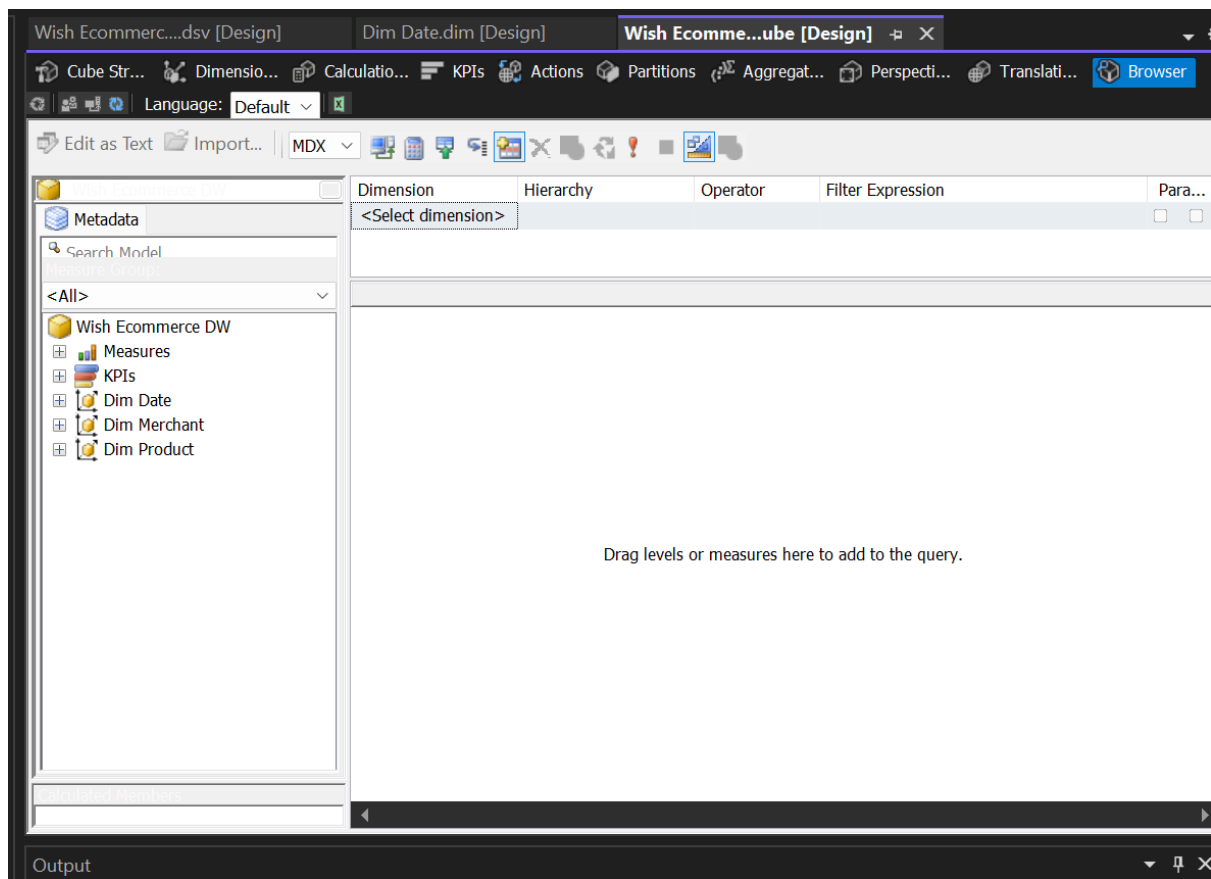


2.5 Cube Deployment

The cube project properties were configured to deploy to HEWAGE2003\SSAD (the running SSAS instance). After granting the required SQL Server permissions to the SSAS service account, the deployment completed successfully.



The cube browser was used to verify the deployment by confirming all dimensions and measures were accessible.



Chapter 3: OLAP Operations in Excel

3.1 Connecting Excel to the SSAS Cube

Microsoft Excel was connected to the deployed SSAS cube using Data > Get Data > From Database > From Analysis Services. The server HEWAGE2003\SSAD was entered, and the WishDW_SSAS database with the Wish Ecommerce DW cube was selected. A PivotTable was inserted on the worksheet to perform OLAP operations.

The screenshot shows the Microsoft Excel interface with a PivotTable and the PivotTable Fields task pane. The PivotTable is located in the worksheet and displays data for the year 2020, rolled up to the Year level. The PivotTable Fields task pane is open on the right, showing the Fact Sales cube and the dimensions Dim Date, Dim Merchant, and Dim Product. The Fields list shows the following configuration:

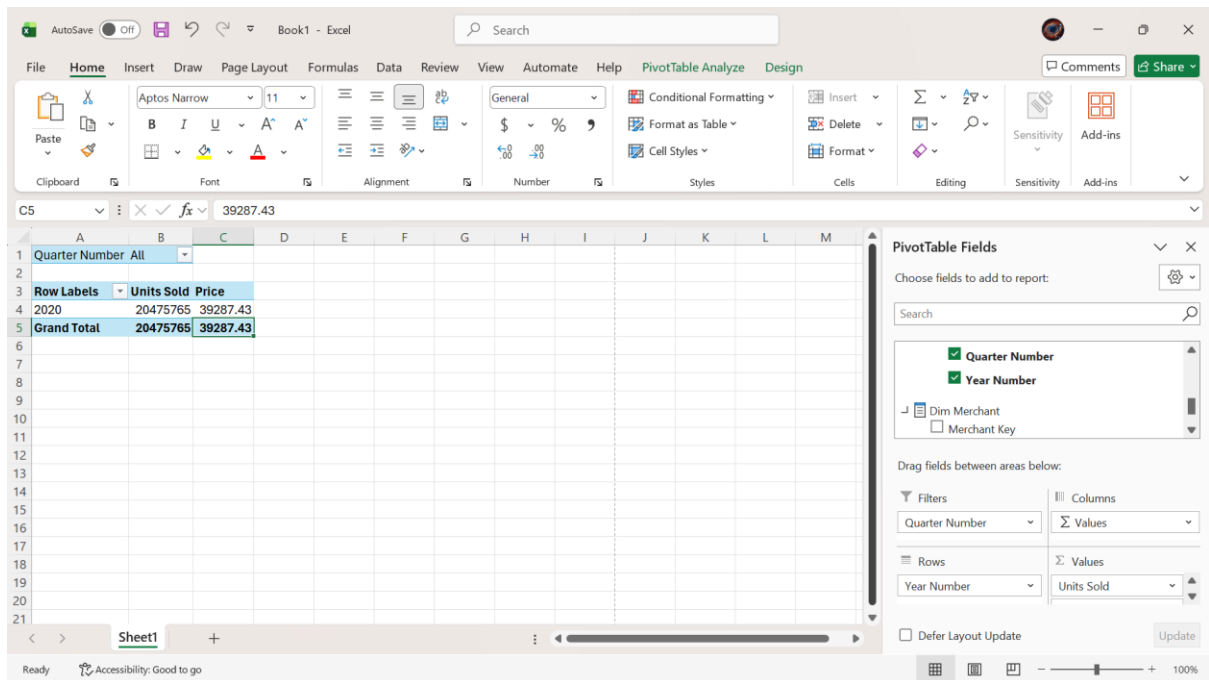
Area	Field
Filters	Merchant Key
Columns	Σ Values
Rows	Year Number
Values	Σ Values

The PivotTable data is as follows:

Column Labels	Units Sold	Price	Total Units Sold	Total Price
Row Labels				
2020	20475765	39287.43	20475765	39287.43
Grand Total	20475765	39287.43	20475765	39287.43

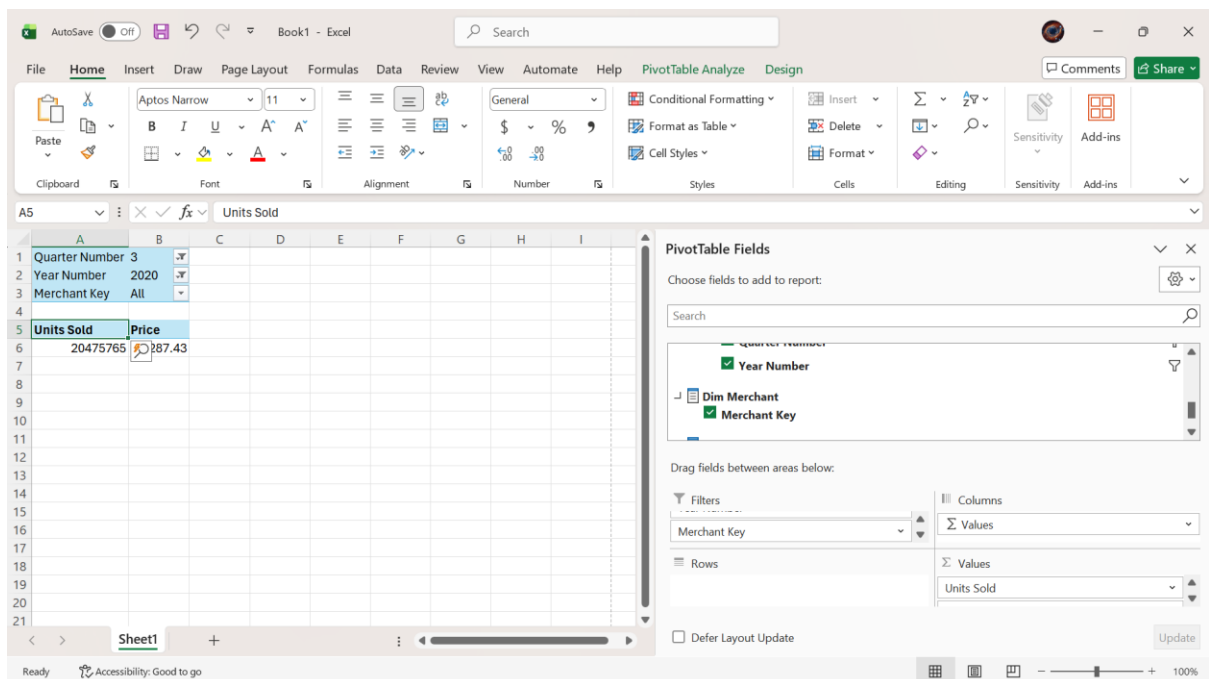
3.2 Roll-Up

Roll-up aggregates data from a lower level of detail to a higher level. YearNumber was placed in Rows and Units Sold and Price were placed in Values. The resulting PivotTable shows total units sold and price rolled up to the Year level for 2020.



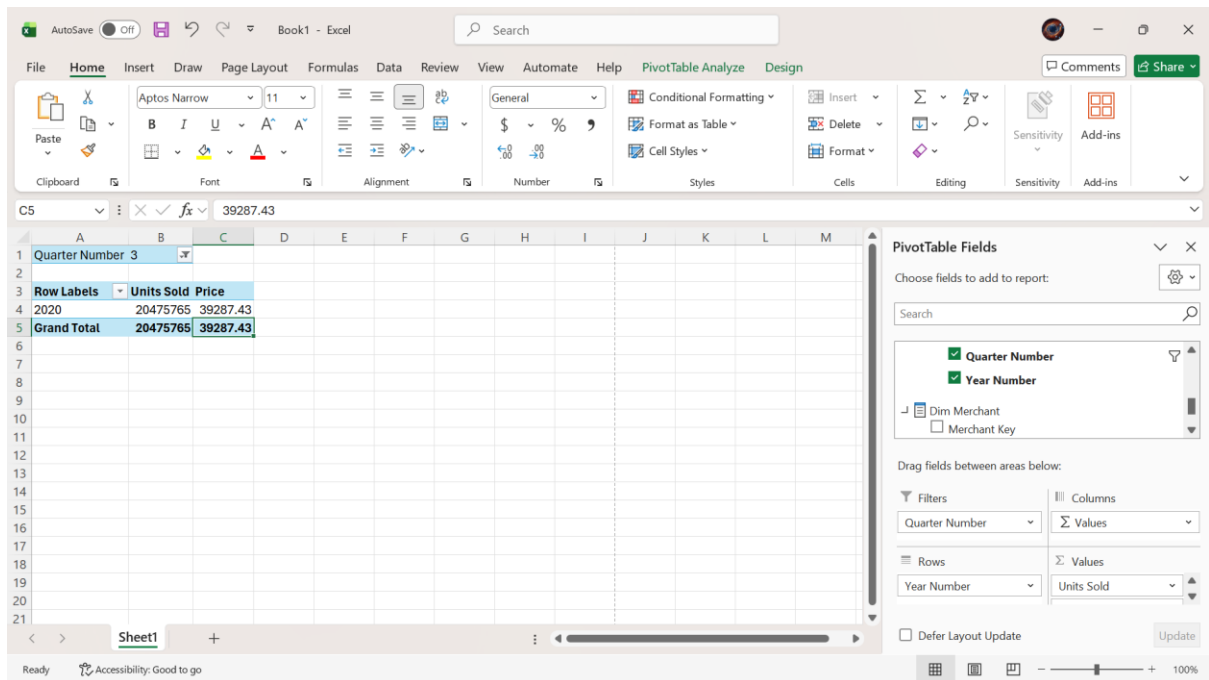
3.3 Drill-Down

Drill-down navigates from a higher level of aggregation to a lower, more detailed level. QuarterNumber was added to the Rows area below YearNumber, revealing the breakdown of 2020 data by quarter. This shows the data drilled down from Year level to Quarter level.



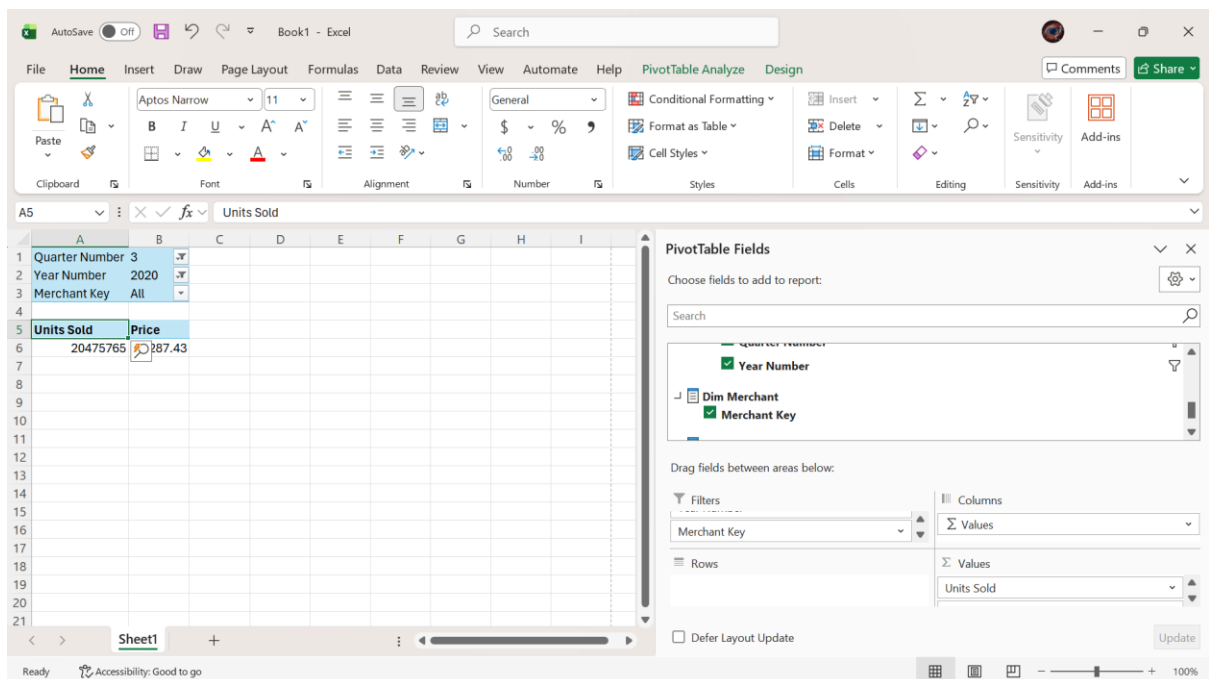
3.4 Slice

Slice filters the cube on a single dimension to show a subset of data. QuarterNumber was moved to the Filters area and set to value 3, effectively slicing the data to show only Q3 2020 results.



3.5 Dice

Dice applies filters on multiple dimensions simultaneously, narrowing data to a specific subcube. Three filters were applied at the same time: Quarter Number = 3, Year Number = 2020, and Merchant Key = All.



3.6 Pivot

Pivot rotates the data axes to present a different perspective. Year Number was placed in Rows and Quarter Number was moved to the Columns area, with Units Sold and Price as Values. This rotation produces a cross-tabular view.

AutoSave Off Book1 - Excel Search

File Home Insert Draw Page Layout Formulas Data Review View Automate Help PivotTable Analyze Design Comments Share

Clipboard Font Alignment Number Styles Cells Editing Sensitivity Add-ins

A5

	A	B	C	D	E	F	G	H
1								
2								
3	Merchant Key	All						
4								
5	Column Labels							
6	Units Sold	Price	Total Units Sold	Total Price				
7	Row Labels	3	3					
8	2020	20475765	39287.43	20475765	39287.43			
9	Grand Total	20475765	39287.43	20475765	39287.43			
10								
11								
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18								
19								
20								
21								

PivotTable Fields

Choose fields to add to report:

Search

☒ Dim Merchant
☒ Merchant Key

☒ Dim Product
☐ Product Key

Drag fields between areas below:

Filters: Merchant Key

Columns: Quarter Number

Rows: Year Number

Values: Price

☐ Defer Layout Update Update

Sheet1

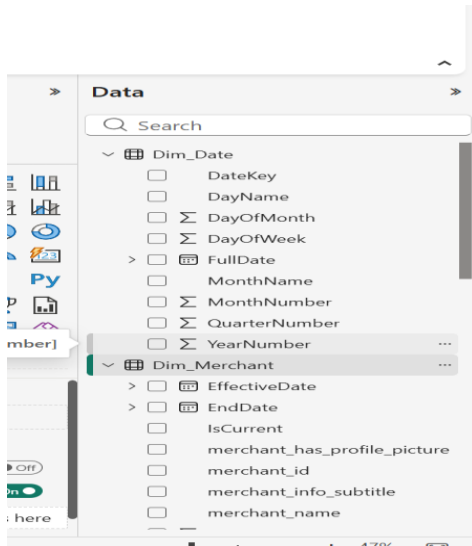
Ready Accessibility: Good to go

100%

Chapter 4: Power BI Reports

4.1 Data Connection

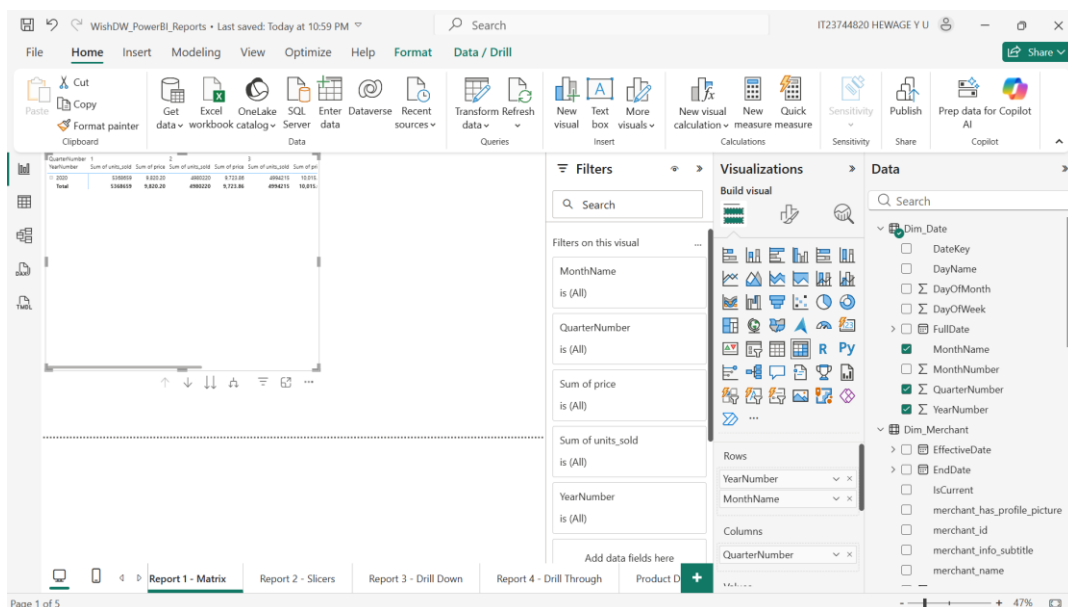
Power BI Desktop was connected to the Wish_Ecommerce_DW SQL Server database on HEWAGE2003\SQLEXPRESS. The four tables Fact_Sales, Dim_Product, Dim_Merchant, and Dim_Date were loaded into the Power BI data model.



The DateKey values in Fact_Sales were distributed across March, June, September, and December 2020 to enable meaningful time-based drill-down analysis. All four reports were developed in WishDW_PowerBI_Reports.pbix and published to Power BI Service using the SLIIT account it23744820@my.sliit.lk.

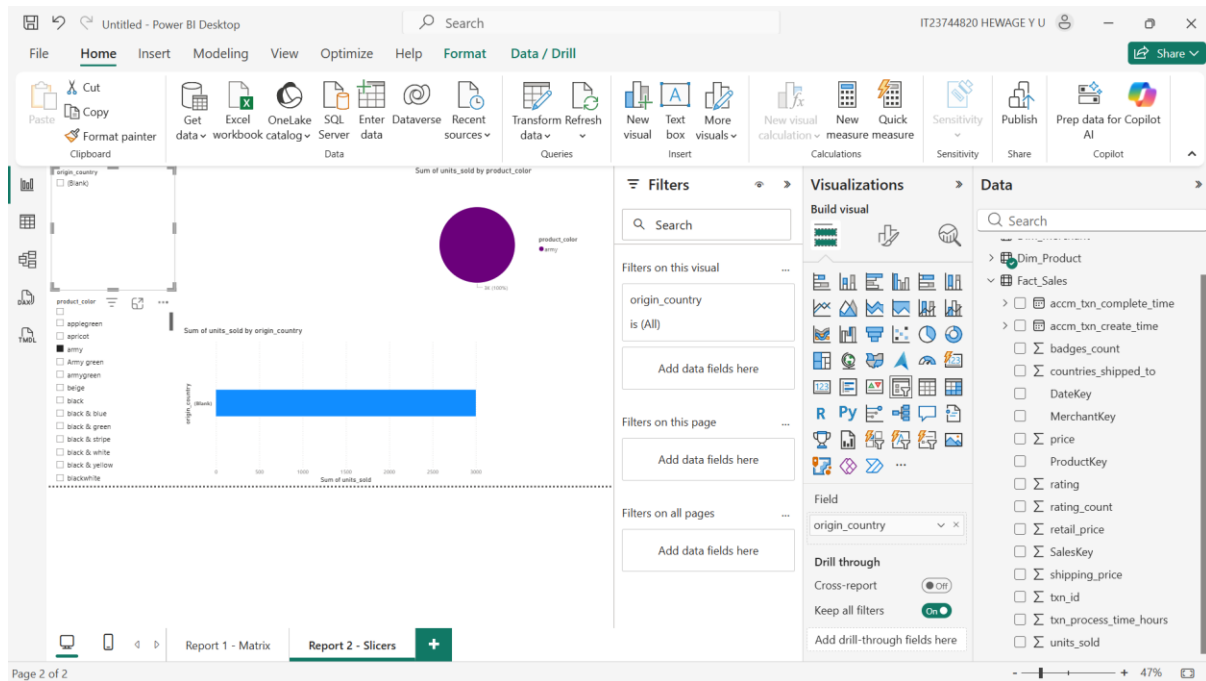
4.2 Report 1: Matrix Visual

Report 1 demonstrates a Matrix visual displaying detailed tabular data with row and column groupings. The Matrix was configured with YearNumber and MonthName in Rows, QuarterNumber in Columns, and Sum of units_sold and Sum of price in Values.



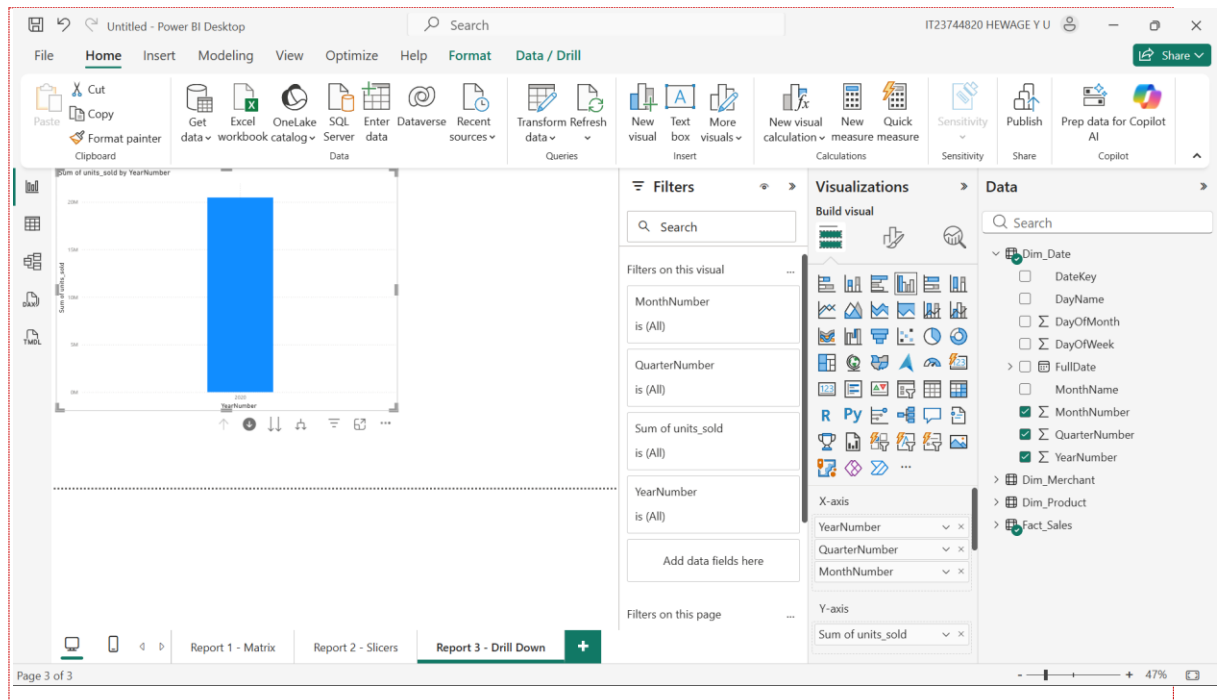
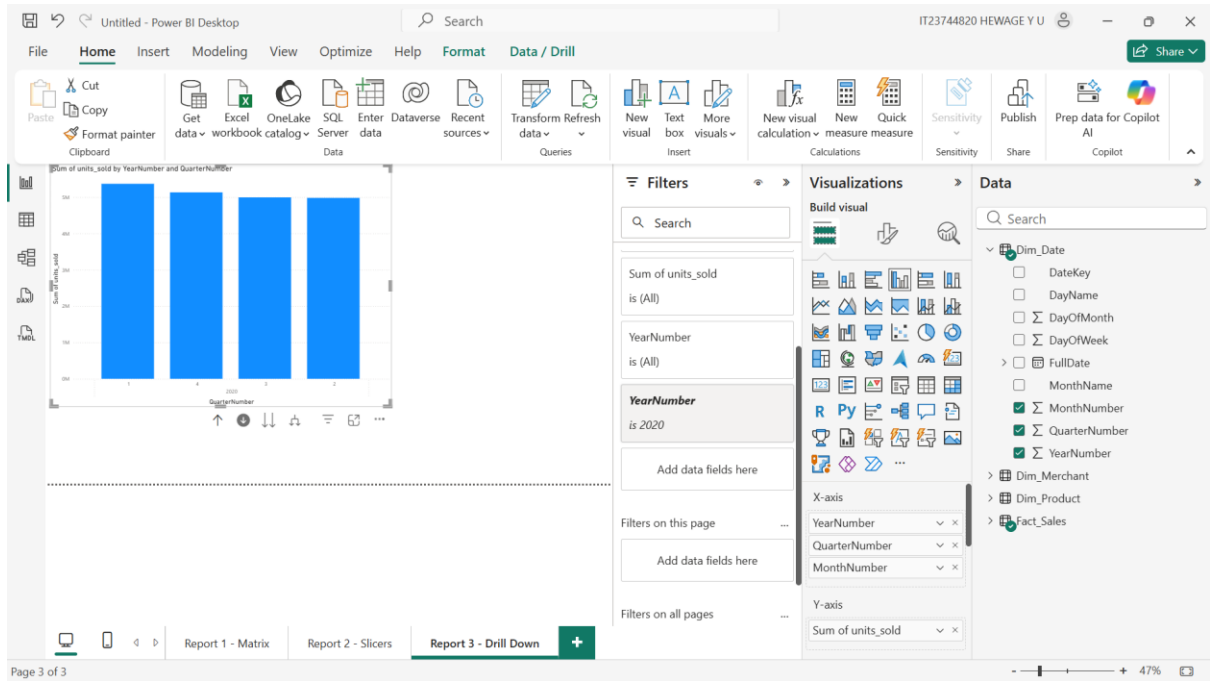
4.3 Report 2: Cascading Slicers

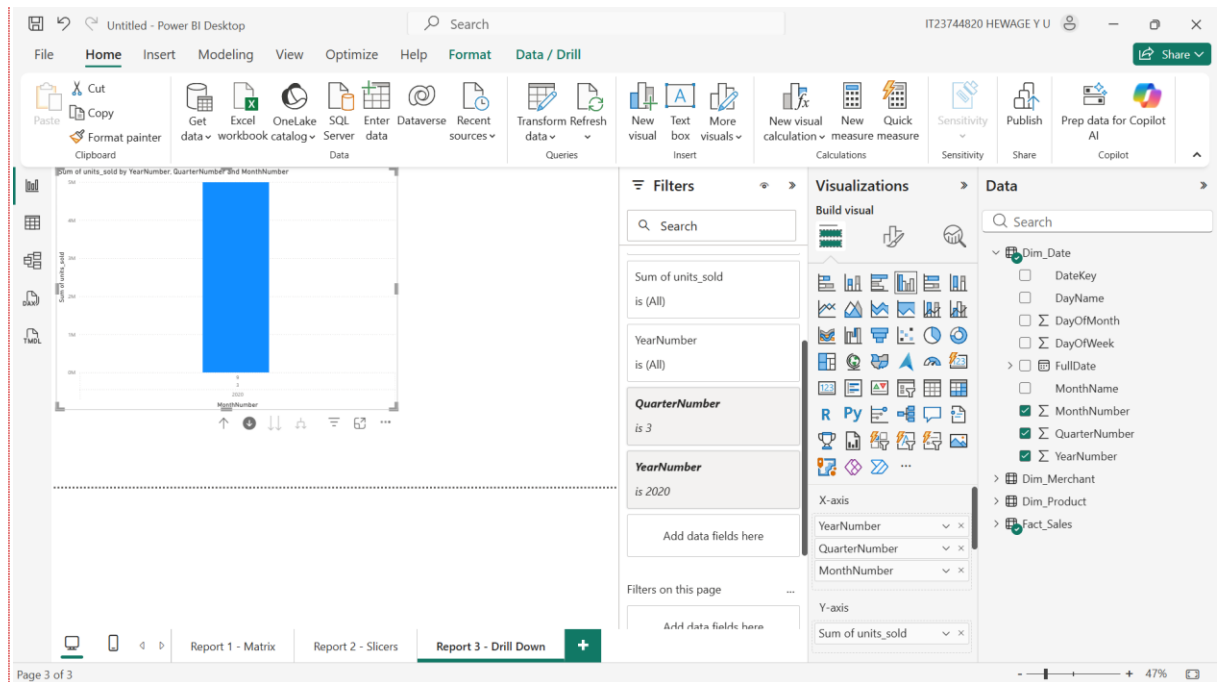
Report 2 demonstrates cascading slicers where selecting a country in Slicer 1 (origin_country) dynamically filters the options in Slicer 2 (product_color), along with a Bar Chart and Pie Chart that update accordingly.



4.4 Report 3: Drill-Down Report

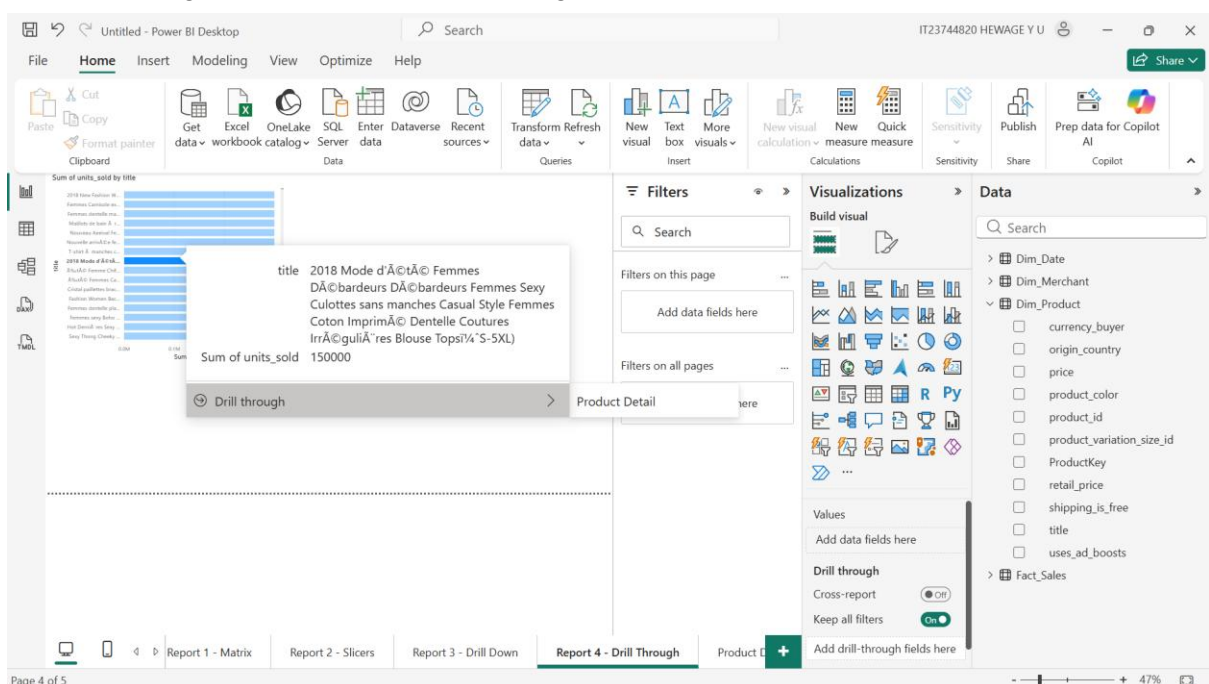
Report 3 demonstrates hierarchical drill-down using a Clustered Column Chart with YearNumber, QuarterNumber, and MonthNumber on the X-axis and Sum of units_sold on the Y-axis.



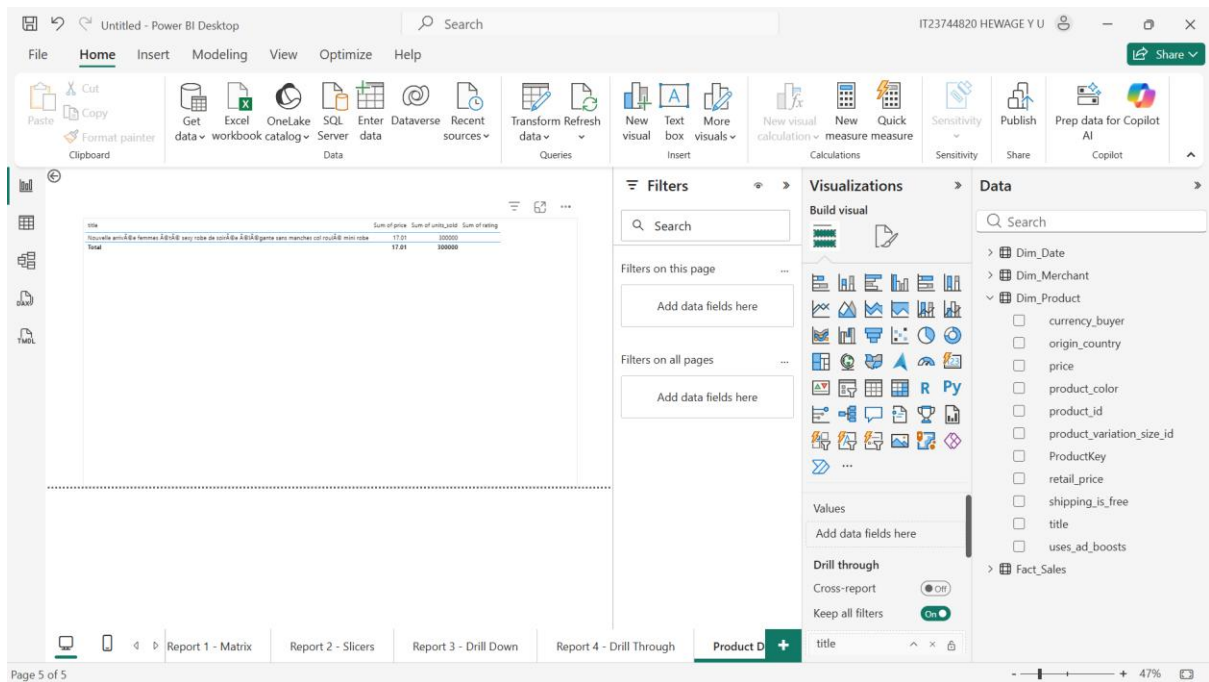


4.5 Report 4: Drill-Through Report

Report 4 demonstrates drill-through navigation. The summary page shows a Clustered Bar Chart of units sold by product title. Right-clicking any product bar reveals the drill-through option to navigate to the Product Detail page.

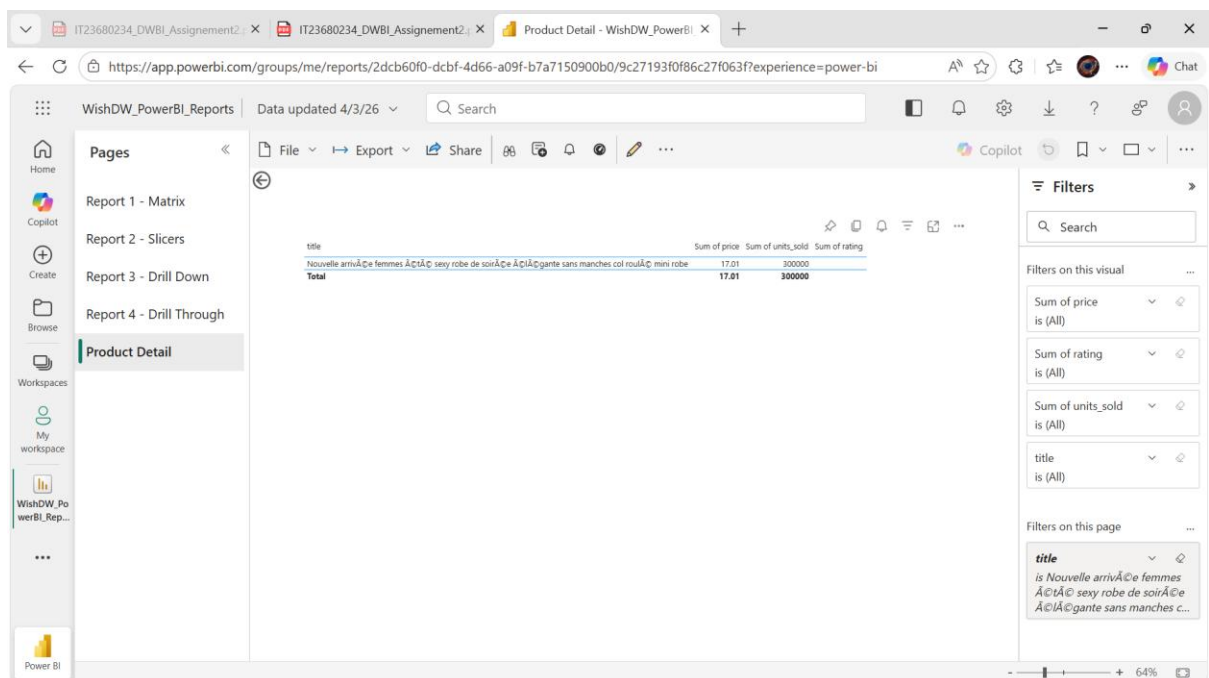


The Product Detail page shows a detailed table for the selected product including its title, price, units sold, and rating. A back arrow button is automatically generated by Power BI to return to the summary page.



4.6 Publishing to Power BI Service

All four reports were published to Power BI Service using the Publish button in Power BI Desktop. The reports were published to My Workspace under the SLIIT organizational tenant using the account it23744820@my.sliit.lk and are accessible online at app.powerbi.com.



Conclusion

This assignment successfully demonstrated the full Business Intelligence stack built on top of the Wish E-Commerce Data Warehouse implemented in Assignment 1.

The SSAS Multidimensional cube was designed with all required components including measures from the Fact_Sales table, three connected dimensions (Dim_Date, Dim_Product, Dim_Merchant), and a Year-Quarter-Month hierarchy implemented in the date dimension. The cube was successfully deployed to SQL Server Analysis Services.

All five OLAP operations (Roll-up, Drill-Down, Slice, Dice, Pivot) were demonstrated using Microsoft Excel connected to the SSAS cube via PivotTables, clearly showing how multi-dimensional analysis can be performed on the data warehouse.

Four Power BI reports were developed and published to Power BI Service, covering Matrix visualization, Cascading Slicers with multiple charts, Drill-Down hierarchical exploration, and Drill-Through navigation to detailed views.

Task	Status	Key Deliverable
Step 1 - Data Source	Complete	Wish_Ecommerce_DW star schema description
Step 2 - SSAS Cube	Complete	WishDW_SSAS cube with Year-Quarter-Month hierarchy
Step 3 - OLAP Operations	Complete	5 operations demonstrated in Excel PivotTable
Step 4 - Power BI Reports	Complete	4 reports published to Power BI Service